LARKEN

DISK DRIVE

FOR TS2068

SPECTRUM EMULATOR COMPATIBLE

OPERATING MANUAL VERSION L3 & L3F

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INTRODUCTION to LKDOS

The LARKEN disk system for the Timex TS2068 is fully Spectrum compatible and also supports AROS or LROS cartridge ROMs. LKDOS supports all token keywords - CAT, ERASE, LOAD, SAVE, MERGE, OPEN #, CLOSE # as well as GOTO and PRINT that were intended for use with external mass storage devices. FORMAT and MOVE are supplied on disk as programs that run in RAM. The information in this manual pertains mainly to users of the pure LARKEN L3/L3F System using a LARKEN RAMDISK and a LARKEN V2 Spectrum EPROM in the LROS socket of the DOS board plugged into the Dock port. Users of ROMswitch and other non-LARKEN hardware may find that some features explained herein do not work in their particular system.

The way these commands are implemented by the LARKEN system is to precede them with RAND USR 100:

```
e.g.: RANDOMIZE USR 100: CAT "",
```

[See NOTES before using PRINT #4:] The PRINT #4: command can be used instead of RAND USR 100: for easier reading and a shorter command line. You must first OPEN stream 4 to the disk drive to use the PRINT #4: command with the initialization command: RAND USR 100: OPEN #4,"dd"

You can now precede all LKDOS and Extended BASIC commands with PRINT #4:

```
e.g.: PRINT #4: LOAD "filename.ex"
```

If the PRINT #4: command is used before the command has been initialized, error report code O Invalid stream will result.

LKDOS File Names

The only other difference between an LKDOS command and a standard cassette command is the file name.

LKDOS uses a file name that contains a program name of up to 6 characters followed by a two character 'extension'. A period (.) separates the program name from the extension. The first letter of the extension tells LKDOS what type of file it is. It must be an A, B or C.

'A' for Array **'B' for BASIC** **'C' for Code**

The extension must be only two characters long. The second letter of the extension in all filenames except string arrays can be any character. The second character of a string array must be a '$'.

The only character that can't be used in a file name is a '^' which is used as a 'wild card' character for use in CAT searches. Some examples of file names:

"Progrm.B1"	A BASIC program
"zeus.Cx"	A Code file
"Names.A$"	A String Array
"Numbrs.A1"	A Numeric Array

SAVE COMMANDS

Before you can SAVE your programs and data to disk, the disk must first be formatted. (See section on formatting.) Any formula or expression in a command can be used.

```
e.g.: PRINT #4: SAVE a$(TO 6)+".CT" CODE Start,End-Start
```

This example takes the first 6 characters of an INPUTed A$ and adds the extension .CT to create a Tasword 2 filename.

All variations of cassette commands are supported.

```
PRINT #4: SAVE "Prog.B1"                    - BASIC program
        : SAVE "Prog.B1" LINE 100           - BASIC Autorun
        : SAVE "Prog.C1" CODE start,length  - Bytes SAVE
        : SAVE "Prog.C$" SCREEN$            - Screen SAVE
        : SAVE "Prog.A1" DATA ()            - Numeric Array
        : SAVE "Prog.A$" DATA $()           - String Array
```

Make sure the write protect notch on the disk is not covered by a protect sticker and that the disk is in the drive properly with the door closed before attempting a SAVE.

LOAD
Similar to the SAVE routines, all cassette LOAD commands are supported.

A special feature in the LOAD command will allow your BASIC program to continue even when a 'File Not Found' error has occurred. If you POKE 23728,100 before doing a LOAD command, your program will not stop with error T when LKDOS can't find the file. LKDOS will instead print "NO FILE" on the screen in the current print position, put 101 in 23728 to indicate it couldn't find the file and continue executing the BASIC program. This feature was added for programs such as word processors that LOAD and SAVE text or manage files from within.

The ON-ERRor command does not recognize LKDOS errors.

MERGE
The MERGE command differs from the cassette MERGE in a few ways. Programs aren't automatically stopped when merged as they are using the cassette merge routine. This permits 'BASIC overlays' which means that a BASIC program can be much larger than what can normally be LOADed into the computer at one time. You can have parts of a program merged into your main controlling program whenever necessary.

Two rules determine what line number the program will continue at after a MERGE command.

- If a running BASIC program merges another program into memory, the first BASIC program will continue and have control over the merged program ignoring the merged program's autorun line number.
- If a program is merged from an immediate command (not running), the merged program will start at its autorun line number. If it is not an autorun program it will not run.

Due to the fact that programs are stored on the disk in blocks, the maximum size of any line in a program to be merged is 1200 bytes. Merging from disk is a complex process and there will be noticeable difference in the speed of a merge compared to a LOAD. The drive may turn on and off during the merge.

The MERGE command doesn't merge program variables, only the program lines. This makes BASIC overlay programs easier to control.

CATalog
The CAT command displays the disk name, all the files on the disk, size of each file in blocks, disk format type and the number of files and free blocks.

The syntax for the CAT in the TS2068 is PRINT #4: CAT "",. The quotes can be closed quotes or they can contain part of a string that you would like to search for in the CATalog.

e.g.: PRINT #4: CAT ".Bz", would print all the files that contained ".Bz"

The '^' is used as a wild card character and can be substituted for any character.

```
e.g.: If a disk contained files progl.B*, Progl.B*, Prog2.BB, Prog2.B*
      and the command PRINT #4: CAT "Prog^.B*" was executed, the CAT
      command would display only - Progl.B*
                                   Prog2.B*
```

The Spectrum does not permit the search function since the syntax is only PRINT #4: CAT The Spectrum will respond, however, with a full CATalog listing if the instruction PRINT #4: CAT"", is executed in a BASIC program line.

To send a copy of the CATalog to the printer you can OPEN stream 2 to the printer and then execute the CAT command.

```
e.g.:            Open #2,"p"  for small printer
      PRINT #4: Open #2,"lp"  for large printer
```

(Note: - use the LKDOS CLOSE command to CLOSE #2 only if LKDOS OPENed it. See Extended BASIC commands for uses of the OPEN command.)

ERASE
The ERASE command deletes the file on the disk. The blocks used by the file will be made available again to the list of free blocks. The ERASE syntax requires a comma after the file name:

e.g.: PRINT #4: ERASE "File",

GOTO
The GOTO command is used to select which disk drive is to be used for the next disk commands. The LARKEN disk interface will support up to 4 disk drives, numbered 0 to 3. PRINT #4: GOTO 4 will select the RAMDISK.

The selected drive will stay selected until another GOTO command is executed or the system is turned off.

Drive 0 is selected during power up. Pressing 'j' on power up will select the RAMDISK instead of Drive 0. Pressing 'j' and ENTER on power up will boot the AUTOSTART program on the RAMDISK.

PRINT
The PRINT #4: PRINT "File" displays the file to screen directly from disk without altering your program memory. It can be useful to search a disk for a program or even to examine an NMI SAVE to see the contents of memory. The listing of BASIC programs will not print line numbers, but word processor and text files are fairly accurate. This command was inspired by the CPM 'type' command.

LPRINT
The PRINT #4: LPRINT "file" command is basically the same as the PRINT command except the file is sent to the printer instead of the screen. If the file contains printer control codes (escape codes) you should PRINT #4: POKE 16098,3 and also POKE 16093,32 to disable any filtering by the printer driver and LPRINT command.

NEW
Entering the command PRINT #4: NEW will cause LKDOS to do a warm start. This can be used to re-LOAD an AUTOSTART program from the current drive. A warm start doesn't change any of the settings in the cartridge or any selected banks or chunks. If an AUTOSTART is not found on the drive, LKDOS will do a cold start.

RENAME (MOVE)
The PRINT #4: MOVE "oldname","newname" command will rename a file on disk. This is for giving NMI SAVEs a more meaningful name or changing the name of any file. The routine does not check for an extension (.B or .C etc.) so use it wisely. You can rename an NMI SAVE with a '.B' extension to allow it to be LOADed without using the CODE token.

RENUMBER
Users of the LARKEN V2 Spectrum EPROM can execute PRINT #10 to renumber a BASIC program by 10 starting with line 10 while in the Spectrum mode.

VERIFY
All blocks of the disk are read to see if the disk has any bad blocks. The CRC error followed by the block number will be printed if any blocks can't be read.

USER DEFINED COMMAND
To add additional LKDOS commands, the token 'DATA' can be linked to the DOS. Advanced machine code programmers can add an extra command to LKDOS by using the PRINT #4: DATA a,b,c... command. You must LOAD your new command into an unused area in the LKDOS cartridge RAM and the POKE 8214,(address of command). Your new command will not be affected by NEW.

The LARKEN disk Editor includes documentation on how to use MC with LKDOS and how to interpret BASIC lines.

AUTOSTART PROGRAMS
A BASIC or machine code program named AUTOSTART can be made to autorun at power up. An AUTOSTART program should do little more than lower RAMTOP, save itself and raise RAMTOP again before calling another program.

Since AUTOSTART requires only 150 to 200 free bytes to operate (PRINT FREE to verify this), use CLEAR 27577 to SAVE exactly 5089 bytes (one block). Enter RAND USR 102: RUN or RAND USR 102: GOTO (line number) to cause an NMI type SAVE. After the tune plays, press "d" to SAVE your program from the start of the attribute file to RAMTOP. This program will autorun when the computer is turned on with the ENTER key pressed. The memory above RAMTOP will be clear, no UDGs. These can be LOADed as a code file if necessary. The line at the bottom of the screen is actually all the Z80 registers, interrupt information and stack pointers. Sample AUTOSTART programs are in the back of this manual.

Though an AUTOSTART SAVEd in the TS2068 mode will not run in the Spectrum mode and vice versa, a pure full featured LARKEN system can switch between systems from any drive using the LARKEN NEW command if the code below is implemented.

Some Spectrum emulators need to be turned on by the command OUT 244,3. To make this type of emulator able to autorun a program, you need to add a small machine code routine. The best place for this routine is in the printer buffer at location 23300.

Assembly Language	Decimal equivalent
CALL 102 (decimal)	205, 102, 0
LD A,3	62, 3
OUT (244),A	211, 244
RET	201

After this program has been LOADed, you can SAVE your autorun program by executing the command RAND USR 23300: RUN or RAND USR 23300: GOTO (line number).

RAND USE 100: NEW will now change from the TS2068 mode to the Spectrum mode from any drive. Replace the 3 with a 0 to reverse the effect. This type of AUTOSTART will override the 'k' key request at power up.

PUSH BUTTON (NMI) SAVEs
Located on the top left corner of the disk interface is a push button that will SAVE any program running in memory to disk. It is known as a 'snap shot' or NMI SAVE push button. The button triggers a Non-Maskable Interrupt (NMI) in the Z80 microprocessor. LKDOS software in that area SAVEs the program along with all registers and stack pointers. (The line on the bottom of the screen is actually these registers).

When the NMI push button is triggered the computer will play a tune and then pause. The entire program and screen can be SAVEd by pressing keys 1 to 5 which will name the file "NMI-S(1-5).CM". If you do not need to SAVE the screen and want to conserve disk space as well, you can press Caps Shift while pressing 1 to 5 and this will do a shorter NMI SAVE that is also compatible with Versions 1 and 2 of LKDOS.

The display file can be SAVEd as "SCREEN.CM" by pressing "s".

Pressing the "a" key will attempt to stop the program by a RST 8 (RESTART 8H).

Pressing ENTER will CONTinue the program should you decide not to do a SAVE.

To LOAD the NMI SAVEd program, you need to LOAD it as a CODE file.

e.g. : PRINT #4: LOAD "NMI-S4.CM" CODE

USER DEFINED NMI ROUTINE
Advanced programmers can LOAD their own NMI routine and have it run when they press the 'f' key after pressing the NMI button. Your code should be LOADed into the cartridge in an unused location and 8214 should be POKEd with the start address of the routine. The routine should end with a RET and should not do any Sinclair ROM calls or disturb the registers stored at the bottom of the display file. This routine could be a screen copy, disassembler, monitor, etc. Code can be LOADed into the cartridge using standard LKDOS LOAD code commands.

SEQUENTIAL FILES
A sequential file is a file that is read from or written to, one character at a time. In the same way you print to the screen or the printer by sending one character at a time, you can now send characters or data to a file on disk.

The TS2068 uses streams 0 to 15 to send and receive data to the screen or printer or from the keyboard. Streams 2 to 15 can be also used for LKDOS sequential files.

USES of SEQUENTIAL FILES
Sequential files are normally used for string data but are also very handy for transferring text and data between different programs.

By opening the sequential file to stream 3 (the printer stream) you can 'print' your data to a file instead of printing it to the printer. You could then, for example, LOAD the file into a word processor for modification or read the data back a bit at a time into a program.

You can also convert BASIC listings into ASCII text by listing them to a file, or print the output from an assembler or disassembler to a file.

You can also read array, BASIC or code files using INKEY$.

Using a modem you can send and receive very large files (larger than 100K) directly to and from disk, eliminating the need for a large memory buffer.

(Note - The TS2068 uses the command PRINT # to send data to a stream. e.g.: PRINT #2; "the brown fox". Do not confuse this with the 'PRINT #4:' that is used to precede LKDOS commands. In the following examples RAND USR 100: is used instead of PRINT #4 to precede LKDOS commands for clarity.)

WRITING to Sequential Files
To send data to a file, you must OPEN it as OUTput by entering the command RAND USR 100: OPEN #S, "filename OUT"

(S = Stream 2-15, the file name can be anything up to 9 characters, OUT or IN is the Sinclair token on the 'i' or 'o' keys. There must be one space between the filename and the OUT or IN token)

After the file has been opened you send data to it using the PRINT # command.

e.g.: PRINT #7; "HELLO"

The data sent should be separated into 'records' by using a comma or a return character (13) as a separator. This will allow the data to be read back a record at a time.

After all data has been sent to the file, you must CLOSE the file.

e.g.: RAND USR 100: CLOSE #7

Here is a program to demonstrate outputting to a file.

```
10 RANDOMIZE USR 100: OPEN #5; "Test.Cf OUT"
20 FOR A=0 TO 10
30 PRINT #5; "RECORD -";A
40 NEXT A
50 RANDOMIZE USR 100: CLOSE #5
```

Note that only OPEN and CLOSE commands are preceded by the LKDOS RAND USR 100: and not the PRINT command.

You can also send a program listing to a sequential file by using LIST #(num.) and if the file is open to stream 3 (printer stream) you can use LPRINT or LLIST.

READING from a File
To read data back from a file, you must open it as INput (IN). Again, there must be one space between the filename and the IN token.

e.g.: RANDOMIZE USR 100: OPEN #7,"filename IN"

Then you can use the commands INPUT # or INKEY$ # to read back the data in the file.
e.g.: INPUT #7;a$; or INPUT #5; LINE a$ or LET a$=INKEY$#7

This program will read back the file and print it.

```
10 RANDOMIZE USR 100: OPEN #5,"Test.Cf IN"
20 INPUT #5;A$;      (or INPUT #5; LINE A$)      (or LET A$=INKEY$#5)
30 PRINT A$
40 PAUSE 30 (delay to show data is read in by records)
50 GOTO 20
```

(note that INPUT or INKEY$ are not preceded by RAND USR 100)

INKEY$ # reads only one character at a time, but INPUT # reads an entire record.

If you read a file right to the end, you will get an End Of File (EOF) error and the file will be closed automatically.

If you don't read to the end of the file you must close it before any other LKDOS commands can be used.

LKDOS will send a CHR$ 255 as its last character. If the file is read further you will get the EOF error.

-------- Notes on Sequential Files --------

- When a file is open, the only LKDOS command allowed is CLOSE. Be sure to use RAND USR 100: CLOSE # and not just CLOSE #.
- INPUT works very much like a normal keyboard input. It will not accept some characters such as quotes '"' or unprintable characters. You can use INPUT LINE to read in quotes. Also, there will be a 'click' noise for every character read using INPUT. However, INKEY$ will read in any character.
- You can have multiple variables in an INPUT statement but they should only be separated by a semicolon.
  e.g.: INPUT a$;b$;
- Stream 2 is used by the TS2068 to print to the main screen. If you open a file to stream 2 for OUTput, you can use ordinary PRINT commands without specifying a stream number. This is tricky to use as no printing is send to the screen.
- Do not change the disk in the drive when a file is open.
- PRINT #4: POKE 16098,3 will turn off de-Tokenizing in the sequential file OUTput routine. This lets you send all characters, 0-255, to the file without the LKDOS treating these as Timex Tokens. This means you can use sequential files to store machine code files. POKE 16098,0 to return the de-Tokenizing routine to normal.
- The file name for a sequential file does not need an extension (e.g.: .B or .C). This allows a lot of freedom but there are limitations, such as, you can not name it with a .B in the file name and LOAD as BASIC. If the file is not too big, you can LOAD it as code.
- Only one file can be open at once. Also, data cannot be added to a file once it is closed. (These restrictions do not apply to the RAM based version of Sequential Files available on disk)

EXTENDED BASIC COMMANDS
These commands consist of extended graphic, additional PRINT and I/O device streams and up to 3 scrolling windows on the screen capable of proportional spaced characters, 42 to over 80 characters per line.

GRAPHIC COMMANDS

- **PRINT #4: DRAW width,heigth,pattern** This is a versatile box fill or CLEAR command that starts at the last plot command that fills in the selected pattern 0 to 9 (or 10 = user defined pattern that consists of 8 bytes starting at address 23540)

- **PRINT #4: CIRCLE x,y,pattern** This is a region filling command that fills from right to left, selected blank areas of the screen in the patterns described above. It will fill a line until it finds a set pixel. x,y sets the starting position.

- **PRINT #4: INK** x Immediately changes INK color on screen.

- **PRINT #4: PAPER** x Immediately changes PAPER color on screen.

- **PRINT #4: POKE** a,b This does a double POKE for 16 bit numbers. b can be 0 to 65535. This command can be used to POKE into the LKDOS cartridge if the address is below 16384. This can be used to change parameters for printers, windows or files.

PEEKing into the Cartridge
The method for PEEKing into the LKDOS cartridge EPROM or RAM is to first PRINT #4: POKE 8200,(address to be PEEKed). Then USR 110 will return the PEEKed value.

e.g.: this program will print the first 10 bytes of the LKDOS EPROM.

```
10 FOR A = 0 TO 10
20 PRINT #4: POKE 8200,A
30 LET B = USR 110: PRINT A, B
40 NEXT A
```

(do not PEEK or POKE addresses 96 to 111 or the cartridge will crash)

WINDOW COMMANDS AND STREAMS

- **PRINT #4: OPEN #n,"device"** This attaches a stream to a display device. (n can be stream 2 to 15.) The device can be "w0", "w1", "w2" (window 0 to 2). (See other uses of the OPEN command for sequential files and a large printer elsewhere in this manual.)

- **PRINT #4: CLOSE #n** (n can be 2 to 15). This is used for closing streams opened with the above command. Note - never use the Sinclair CLOSE command (not preceded by LKDOS switch) to close a stream opened by LKDOS.

- **PRINT #4: INPUT #(window),top,left pos,right pos,bottom pos**
This command defines a window size, position and color. "window" is 0-2, other parameters are standard print positions in absolute numbers (0 to 21 or 31).

e.g.: PRINT #4: INPUT #1,0,10,20,8

This would define window #1 to have its upper left corner at print pos 0,10 and the window would be 10 units wide (20 - 10). The bottom of the window would be at the 8th line. The screen colors that are being used when the window is defined will stay with that window.

- **PRINT #4: CLEAR w** This will clear the window (0-2) and restore the print position to the top of the window. It will also re-color the window.

WINDOW CHARACTER SETS

When you turn on your computer the LKDOS LOADs the TS2068 character set into the cartridge for use with the windows. The character set LOADs into the cartridge at address 15200 and is 768 bytes long. You can also LOAD in 42, 51, 64 or proportional spaced character sets for use with the windows.

e.g.: PRINT #4: LOAD "Chars.C1" CODE 15200,768 would load in the 42/51 proportional character set from the LARKEN System Disk.

The first byte in each character in the character set determines the width of the character. The number of pixels cleared (not set) in the first byte (top pixel row) starting from the left sets the width of the character. No pixels set would indicate a full 8 pixel wide character.

e.g.: this is the character pattern for a 'Y' character that is 5 pixels wide. (You could print 51 of these across the screen)

```
00000111        - This sets a character width of 5 pixels
10001000        - Character data
01010000        - Character data
00100000        - Character data
00100000        - Character data
00100000        - Character data
00100000        - Character data
00000000        - Character data
```

USING WINDOWS

To set up a window you need to OPEN a stream to a window and then define the size and position of the window with the PRINT #4: INPUT command.

To use the window you can use the standard TS2068/Spectrum commands (not preceded by the LKDOS switch) such as PRINT #n or LIST #n

e.g.: If stream 15 has been opened to a window then:
PRINT #15;"TEST" would print to that window, or
LIST #15 would list your program to that window.

The computer uses stream 2 as the stream for standard PRINT commands to the main screen. You can OPEN stream 2 to a window or printer. All standard PRINT or LIST requests will now be directed to the selected device.

There is a Window demo in the back of this manual.

REMARK about STREAMS & MACHINE CODE

When any RAND USR 100: OPEN # or PRINT #4: OPEN # command is used, 50 bytes (total) of memory is added to the stream table. This is not reclaimed by the PRINT #4: CLOSE command and may cause a problem for a program that has machine code in a REM statement.

Instead of calling a USR routine at 26715, 5 higher than the normal beginning of BASIC most often at address 26710, use PEEK 23635+256*PEEK23636 to find the new address for the beginning of BASIC and add 5 to locate the beginning of the USR routine. You may find that the USR routine works equally well in both the TS2068 and Spectrum modes using this method.

LARGE PRINTER DRIVER
The command PRINT #4: OPEN #3,"lp" will allow you to use the standard LPRINT and LLIST commands for your large printer since stream #3 supports both LPRINT and LLIST. You can use both Large Printer and TS2040 at the same time by opening the "lp" to a stream other than 3.

If you OPEN #2,"lp" then PRINT #4: CAT"", will sent the disk directory to the large printer.

The LARGE PRINTER DRIVER can be used with AERCO, modified Tasman B or A+J printer interfaces as well as a user defined printer interface.

You can change line length, left margin setting, and specify line feed and it prints Graphic characters as underlined ASCII characters. TAB and comma "," are also supported.

On power up the default settings are AERCO interface, line length 64, line feed with carriage return and margin of 0.

To change settings you will need to use PRINT #4: POKE Address,Setting. These addresses are in the LKDOS cartridge RAM and are not affected by NEW.

POKE 16090,Maximum width of printed line. This is set at 63 on power up to give a line length of 64.

16092,0 if you don't want a linefeed with carriage return, 10 if LF wanted with CR. This is set at 10 on power up.

16094,Left Margin Setting. This is set at 0 on power up. Do an LPRINT after changing this setting.

16096,0 for AERCO CPI, 1 for Tasman CPI, 2 for A+J CPI, 4 for user defined interface routine. This is set at 0 on power up. For 4 also POKE 8216,(address of the routine).

KEMPSTON COMPATIBLE JOYSTICK AND MOUSE
A Kempston compatible joystick port is on the left side of the disk interface. It is compatible with most software written for the Spectrum.

The joystick port can be read using the command IN 31. Numbers from 0 to 26 represent the joystick positions.

The Commodore C1351 mouse or other compatible units such as the WINNER M3 will work in the Kempston compatible joystick port when the right button is held down during TS2068 power up.

It is a good idea to anchor your joystick/mouse cable before it attaches to the board. This will prevent damage to the disk interface in case you accidentally pull on the joystick/mouse cable.

ADDING a RAMDISK
A 256K RAMDISK can also be controlled by LKDOS as well as 4 floppy drives. The LARKEN 1/4 Meg nonvolatile memory board attaches to the rear of the TS2068 and is available from LARKEN. It acts exactly like the other floppy drives but is very fast. PRINT #4: GOTO 4 selects the RAMDISK. All floppy commands can be used.

You must format the RAMDISK with the PRINT #4: FORMAT "n", command before using it. (n = number of RAM chips on the board). The LARKEN RAMDISK comes with complete documentation.

Hardware experimenters can add a small RAMDISK by adding 32K of RAM in the upper 32K of the dock bank. Users of LKDOS for the AERCO FD-68 can use the AERCO RAM as a RAMDISK.

COPYING DISKS and MOVING FILES
On the disk (or tape) supplied with the system there are programs that are to be LOADed in for copying disks and moving single files.

FORMAT.Bx is used to format disks and to copy entire disks using 1 or 2 drives, x is L for LARKEN (WD1770), F for FAST (WD1772), O for Oliger, A for AERCO and R for RAMEX.

MOVE.B1 is a program for moving 1 or more files using 1 or more drives. It is very easy to use. If you have only one drive, enter drive 0 for both source and destination.

FORMATTING
A floppy disk, unlike a cassette, must be formatted before data can be saved. Formatting initializes all tracks and installs a track map for up to 160 tracks, directory for up to 100 filenames, and disk parameters on track 0. Any errors will be reported as CRC errors followed by a number indicating the track number. A good format should have no errors reported.

The FORMAT command is supplied on cassette. Load the cassette, put a disk in drive 0 and close the door. The first thing to enter is the Disk Name. Usually the Disk Name is a short name (less than 1 line) but LKDOS allows the Name to be up to 1000 Characters. This can be used for a disk description, etc. The delete key (shift 0) can be used to correct mistakes but the editing is limited. If you make a big mistake you can BREAK the program and RUN it again.

To end the entry of the disk name press STOP (symbol shift A). The drive number can be drive 0 to 3. Next is the head step rate referred to as head speed in LKDOS programs and literature. It must be 6, 12, 20 or 30 milliseconds. Most modern drives can run at a 6 ms head speed. Some older single sided drives such as the SA400 Shugart run at a slow 20 ms. (Newer disk interfaces use a 1772-02 controller chip for the faster 3.5" drives. The format program supplied will have a head speed selection of 2, 3, 5 and 6 ms.)

The number of sides should be 1 or 2. (Single or double sided)

The next parameter to enter is the number of tracks. This number should be 35, 40, or 80. Most drives are 40 track. A 40 track drive is a double density drive and an 80 track drive is a Quad density drive.

Type "y" if the drive is ready to format the disk. The drive should turn on and step from track to track until the format is completed.

The format program can now be saved onto the disk for future use.

Break the program and type RAND USR 100: CAT "", You should see a directory of the disk with all the disk info displayed. The term BLOCK refers to one track. The total Blocks on a disk is the (number of sides) X (the number of tracks).

The DOS numbers the blocks starting from track 0 side 0. Block 1 is track 0 side 1. Block 2 is track 1 side 0. Block 3 is track 1 side 1 and so on.

The format routine creates a track map on the directory track (block 0) that corresponds to the number of tracks and number of sides that were entered.

When you format a single sided disk, the format routine creates a track map on the disk that has all side 1 tracks set as in use.

Because the DOS reads tracks from side to side, you cannot read a double sided disk with 1 single sided drive. Also, because of track density, disks formatted on an 80 track drive can not be read from 35 or 40 track drives.

You can, however, read single sided disks with a double sided drive. So if you have a double sided drive and you would like to transfer files to someone with a single sided drive, you can format your disk as single sided.

LKDOS ERROR REPORT CODES
In addition to a number of Sinclair error report codes used by LKDOS, a number of extra error report codes are added.

The On-ERRor command in the TS2068 does not recognize these extra error codes.

S - Protect Error	: Disk cannot be written to. It has a write protect sticker on it.
T - File Not Found	: The requested file is not in the CATalog. This can be overridden. See LOAD command
U - Disk Full	: There is not enough room on the disk to SAVE the file or more than 100 files in CATalog.
V - Wrong File Type	: The file type in the extension doesn't match the syntax of the command. The token CODE or DATA is missing or wrong array type is used.
W - Invalid Command	: The command following the PRINT #4: is not used by LKDOS.
X - CAT Data Error	: The CATalog cannot be read properly so the command is aborted.
Y - File Open	: A sequential file is open and the only command allowed is CLOSE.
Z - Name Exists	: When renaming a file with MOVE this will occur if the new name is already used.
- CRC ERROR (num.)	: A CRC error will be printed on the screen when the DOS cannot read a block after 10 tries. The number following the CRC error is the block number.

NOTES on using the LKDOS System
- Never turn the power to the disk drive on or off when there is a disk in the drive with the door closed. This can glitch a disk and cause loss of data.

- It is a good idea to keep write protect stickers on important disks at all times except when you want to SAVE or delete a file.

- You don't have to open the PRINT #4: command every time you use LKDOS. If, for example, you just want to see the CAT of a disk, it is faster just to type RAND USR 100: CAT "", but in BASIC programs, it is a good practice to use the PRINT #4: command. It uses 3 less bytes than RAND USR 100: and looks neater. Your AUTOSTART program should contain the RAND USR 100: OPEN #4 "dd" near the beginning of the program so that error O - Invalid stream doesn't occur if you feel compelled to use PRINT #4: feature.

- The ultimate RAM saving way to implement LKDOS is to use LET h=CODE "d" instead of RAND USR 100: OPEN #4,"dd" and to use RAND USR h: in place of PRINT #4:.

- Never LOAD or SAVE code over addresses 96 to 111. This will cause the LKDOS cartridge to be turned off during a command and cause a crash. This also applies to PRINT #4: POKE and using PEEK (USR 110) in this area.

- When you remove and install your disk interface from the TS2068 inspect the connector to make sure no pins are bent and also that the locating key is in place. A damaged or missing locating 'key' can cause damage to both the TS2068 and the Disk Interface.

- If you plan on distributing some software on LKDOS formatted disks, using disks formatted as single sided will be compatible on more systems than double sided disks. (48 TPI is most common)

- The LKDOS cartridge is also available for the RAMEX, OLIGER and AERCO disk systems. Your disks are compatible with any of these systems if steprt.B1 is the first program on the disk.

- There are two System 3 versions. L3 uses the WD1770, L3 EPROM and FORMAT.BL while L3F uses the WD1772, L3F EPROM and FORMAT.BF. They can not be mixed or the system will not work properly. L3F serves 3.5" disk drive users best.

[On one LKDOS EPROM modified for "Slow Chip", Larry lifted pin 20 and soldered a jumper from Pin 20 to Pin 14. The chip was a 2764-20. (MBM 2764-30)£ Ed.]

ADDRESSES WITHIN LKDOS THAT CAN BE POKED TO CHANGE PARAMETERS

(using PRINT #4: POKE or PEEKed by USR 110)

8195 - Dvsel - this is the current drive selected. 2=drv0 4=drv1 8=drv2 16=drv3 128=drv4 (RAMDISK) The BASIC statement LET x=INT LN x+(x=16) will return the number of the drive LKDOS is pointing to.

8200 - PEEK - holds the address of the address to be PEEKed by USR 110.

8214 - USERad - holds the address of the user NMI function or the address of the user command (DATA)

8216 - PTdrv - holds the address of the user installed printer interface routine.

8218 - Chars - holds the address of 256 less than the address of Window Character set.

------ **WINDOW POKES** ----------

Each window uses 20 bytes to keep its info. Window 0 (w0) starts at 16000, w1 at 16020 and w2 at 16040. Addresses below are for w0. Add 20 for w1 and 40 for w2. You may have to PEEK the next address and then add 256 X its value to the value being POKEd when using PRINT #4: POKE.

16004 - Wscrol - POKE this with 255 to increase scroll counter just like in the TS2068.

16005 - Xpos - Xpos of the next print pos (0-255). Don't go out of range of window.

16006 - Ypos - Ypos of the next print pos (0-176). Don't go out of range of window.

16014 - Wcol - Ink and Paper color byte for window.

------ **PRINTER AND DE-TOKENIZE POKES** --------

16090 - MWIDE - maximum printer width, 255 maximum for 256 characters.

16092 - LFEED - If 10 then a line feed will be send with each carriage return. If 0 then no line feed will be sent.

16093 - PPAS - if POKEd to 32 then all characters LPRINTed will be sent directly to the printer with no checking. This is useful for sending control characters or graphic info to the printer.
POKE it to 0 to return to de-Tokenizing and filtering.
Re-POKE 'MARG' after poking this one because PRINT #4: POKE does a 2 byte POKE.

16094 - MARG - specifies left margin width. Be sure MARG + MWIDE does not equal more than your printer can print. (80 chars)

16096 - PTYPE - Printer interface type. 0 = AERCO, 1 = Tasman, 2 = A+J, 4 = User defined. POKE PTdrv with address of user code.

16098 - PFLAG - POKE with 3 to defeat de-Tokenizing and filtering in LPRINT "file" command and in printing to a sequential file.
POKE with 0 for restore to normal.

FREE MEMORY in the cartridge for USR routines. Code can be LOADed by the standard LOAD CODE LKDOS command.

13600 - there are 1400 bytes here that are only used when the MERGE command is used.

16100 - There are 283 bytes here free at all times.

LKDOS OPERATION

The LARKEN Disk System provides a cartridge that contains the software for controlling the drive. This is called the Disk Operating System (DOS). On the cartridge is an 8K EPROM mapped in the 0-8K area of the DOS bank and an 8K RAM that is mapped in the 8-16K area of the DOS bank.

The DOS bank is activated when the cartridge senses a USR call (M1 cycle) at addresses 96 to 110 (decimal). For LKDOS commands only the address 100 is used (also 102 is used for SAVEing an AUTOSTART file). The other addresses are used by the DOS as entry points for the PRINT #4: command, PEEK and windows.

The DOS bank is deactivated when a read of write is made to this area of memory.

The command RAND USR 100: OPEN #4, "dd" is used to initialize the PRINT #4: command to point to the LKDOS cartridge. PRINT #4: is easier to type than RAND USR 100: and there is no chance of accidentally typing a wrong USR address. The "dd" means "disk drive".

The LKDOS operating system manages all disk space allocation by using a track map. The map is created by the format program and resides in track 0 along with the CATalog. By using the map LKDOS keeps track of used and unused areas of the disk.

The CATalog can contain up to 100 filenames. The blocks used by each file are kept in the CATalog also.

Full information on LKDOS operation and accessing the DOS from machine code is supplied with the 'LARKEN Disk Editor'.

The rear mounted board is the Disk Drive Interface. This board controls the drive motors and the data transfer to and from the drive(s). It can control up to 4 Shugart compatible drives. (3", 3.5" or 5.25" drives) The disk format used by LKDOS is double density with 10 sectors of 512 bytes per track (or 5120 bytes per track)

On a single sided 40 track drive you will have 200K, on a 40 track double sided drive 400K, and on an 80 track double sided drive 800K per disk.

On each track 5090 bytes are actually used to hold the saved data and the remaining 30 bytes are used by an information header.

Also on the board is a push button that will trigger the NMI line on the Z80 microprocessor. This is used to "capture" any program to disk.

A Kempston joy stick port is also on the disk interface for use with most of the software that is produced for the Spectrum. It can be read using the command IN 31. The Commodore 1351 mouse works in this port.

Memory Map of LKDOS bank, TS2068 Home bank and DOCK bank:

```
0                    16K        32K                              64K

----------------------
| LKDOS CARTRIDGE    |
| 8K EPROM | 8K RAM  |
------------------------------------------------------------------
| Timex Sinclair ROM |                48K RAM                      |
------------------------------------------------------------------
| LROS CARTRIDGES               |  AROS CARTRIDGES OR RAMDISK    |
------------------------------------------------------------------
                                |       RAM DISK BANKS           |
                                |       ""         ""            |
                                |       ""         ""            |
                                |       ""         ""            |
                                ----------------------------------
```

SETUP PROCEDURE

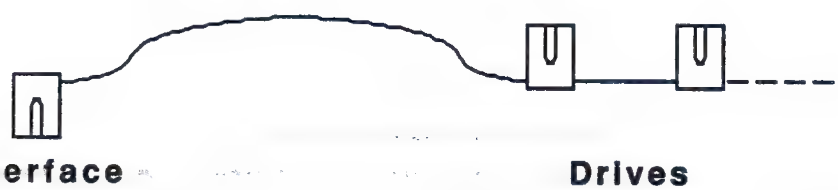
Your new LARKEN disk system contains some chips that can be destroyed by static electricity. This is why the boards are wrapped in aluminum foil during shipping. Observe anti static precautions when handling these boards. Always make sure all power is OFF when installing or removing either of the boards.

In addition to the LARKEN boards, you will need a disk drive(s), a disk drive cable and a disk drive power supply. The disk interface can control any Shugart compatible floppy drive except an 8" drive. The power supply must supply 5 volts and 12 volts. The current rating depends on the number of drives, but a general rule is 1 amp (on both the 5V and 12V lines) per drive.

If you did not purchase the disk drive cable from LARKEN ELECTRONICS, you can make your own using 2 feet of 34 conductor ribbon cable and 2 or more 34 position ribbon cable card edge connectors. The best way to install the connectors onto the cable is to use a vise to squeeze the connector. The connector on the interface end should be mounted opposite to the drive connector(s). If you use more than 1 drive, space the drive connectors 3" apart.

Installing the cable on backwards will not cause any damage, but the drives will turn on when they are powered up.

Interface to Drive Cable



Set the drive select jumpers on each drive as drive 0 to 3 respectively. Make sure all terminator resistor packs are removed from the drives EXCEPT the drive connected closest to the end of the drive cable.

Remove the power from the TS2068. Install the cartridge into the dock port and the interface into the rear of the TS2068. Power up the TS2068 again. Type RAND USR 100: and you should get the error report: - W Invalid Command.

Turn off the TS2068 and connect the drive cable between the interface and drives and power up the drives. The drives should not turn on. If they do, you may have the drive cable reversed. Turn on the TS2068. To determine if drives are connected properly, insert a disk into drive 0. Type RAND USR 100: CAT "", and Drive 0 should turn on (LED and motor) and then after the approximately 15 seconds it should turn off. To test the other drives, use the command RAND USR 100: GOTO drive number (0-3) followed by RAND USR 100: CAT "", to activate the drive.

If you experience X CAT Data errors or other LOADing problems or your drive makes a laud purring sound when turning, the head step rate FORMATed on the disk may be wrong for your particular drive. If LOADing problems occur, try increasing the head speed number. If purring is the problem, decrease the head speed number. If purring does not go away on 3.5" drives, try an L3F system.

RETURNING DAMAGED ITEMS FOR REPAIR

If your LARKEN product is damaged, you can return it to LARKEN for repair. Only return LARKEN Items and do not send disk drive or power supply unless requested so by LARKEN. Sending electronic items from the US to Canada is sometimes a problem so if you must put a value on the package keep the value low (preferably less than $40 or the customs may charge me duty as I have no proof that I am not buying the item). I've had a lot of trouble with this in the past. Also avoid sending by a courier. Send by regular mail.

ADDING an LROS Cartridge chip to the LKDOS Cartridge (Not AERCO FD-68)

Don't attempt this unless you have a proper soldering iron and experience in delicate soldering. LARKEN Electronics will do this modification for $5.

Parts needed - 28 pin IC socket, 74HCT32 IC and #32 insulated wire. Use a low power soldering pencil that is properly grounded.

Temporarily remove the EPROM from the LKDOS cartridge. Bend pins 20 and 22 on the socket outward. Note - all ICs have their pin 1 (dot or notch) towards the center of the cartridge.

Install the 28 pin socket on top of the 8K RAM IC (the 28 pin IC that is soldered to the board). Carefully solder all pins except 20 and 22 (you only need a tiny bit of solder). Carefully remove all leads from the 74HCT32 except pins 1,2,3,7,14. Bend pins 1 and 3 up (fold the lower half back up). Stack this chip on top of the middle 14 pin IC, (pin 1 faces in) and solder the pins 2,7 and 14 to the same pins on the IC beneath.

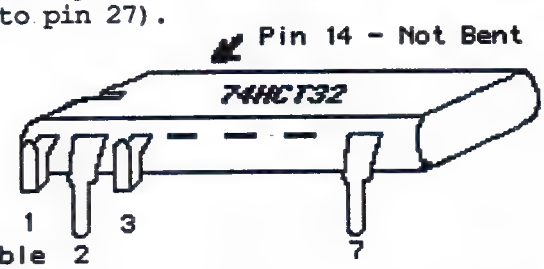
Run a small wire from pin 3 on the 74HCT32 to pins 20 and 22 on the 28 pin socket. Run a small wire from pin 1 on the 74HCT32 to the ROS enable connection near the buss. That's it!

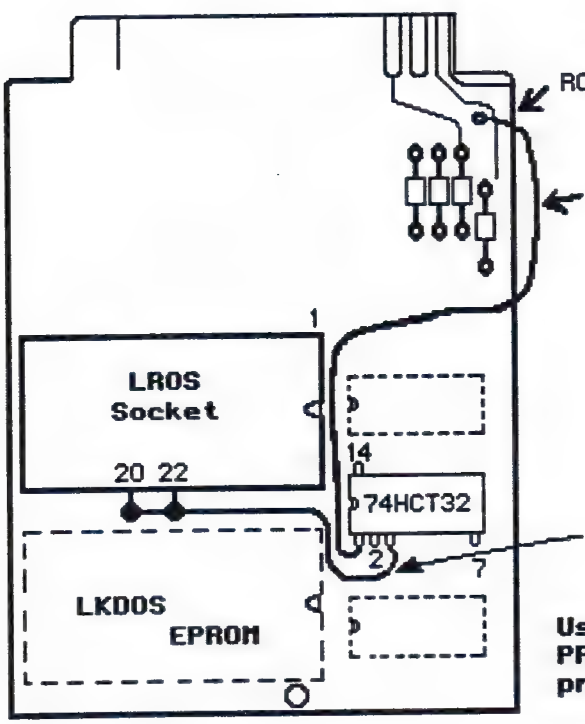
Check for any shorts or solder bridges. Reinstall the LKDOS EPROM and test it in the TS2068 to see if it still works OK.

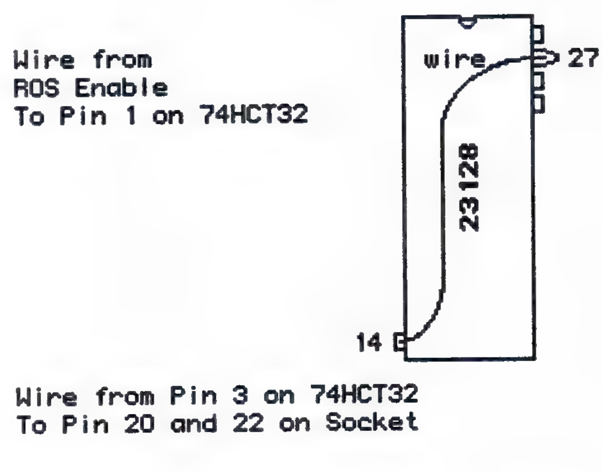
Remove the EPROM from your OS-64 or LARKEN V2 Spectrum emulator and insert it into the socket. If it doesn't work, recheck your work.

If you have a Spectrum ROM chip (not EPROM), part number 23128, it may not work unless you bend pin 27 out (so it doesn't go into the socket) and ground this pin (connect a small wire from pin 14 to pin 27).

The Spectrum ROM can be turned on by the command OUT 244,3 or by pressing the 'k' key on powerup.







Using the LarKen V2 Spectrum EPROM, PRINT #10 will renumber your BASIC program by 10 starting with line 10.

The Spectrum 23128 ROM

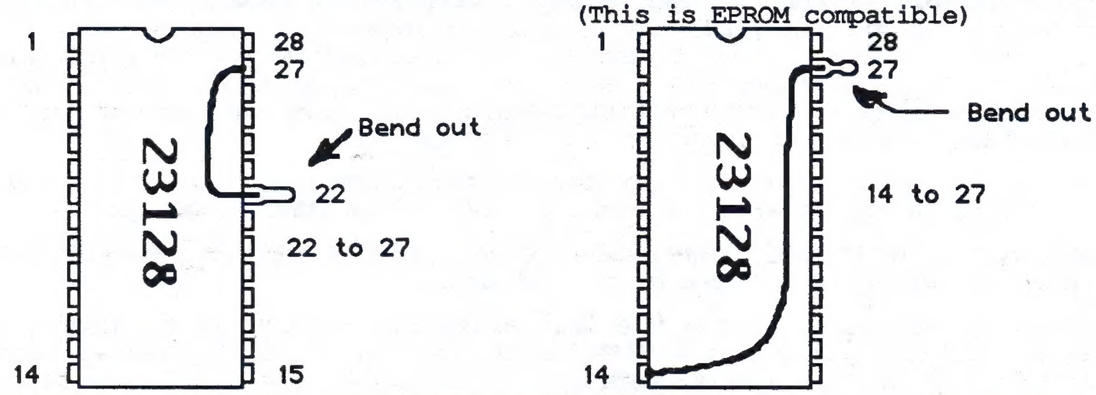
The 23128 ROM (NEC and others) have 3 chip selects which cause problems (it can't be de-selected by LKDOS) when installed in the TS2068 ROM Socket or in the optional EPROM socket on the LKDOS board.

-If you want to remove the TS2068 ROM and install the Spectrum ROM, you must bend out pin 22 so it won't go into the socket and solder a small wire from it to pin 27.

-If you want to install the Spectrum ROM into a socket added to the LKDOS cartridge you must bend out pin 27 so it doesn't go into the socket and solder a small wire from it to pin 14.

Modified for TS2068 ROM Socket

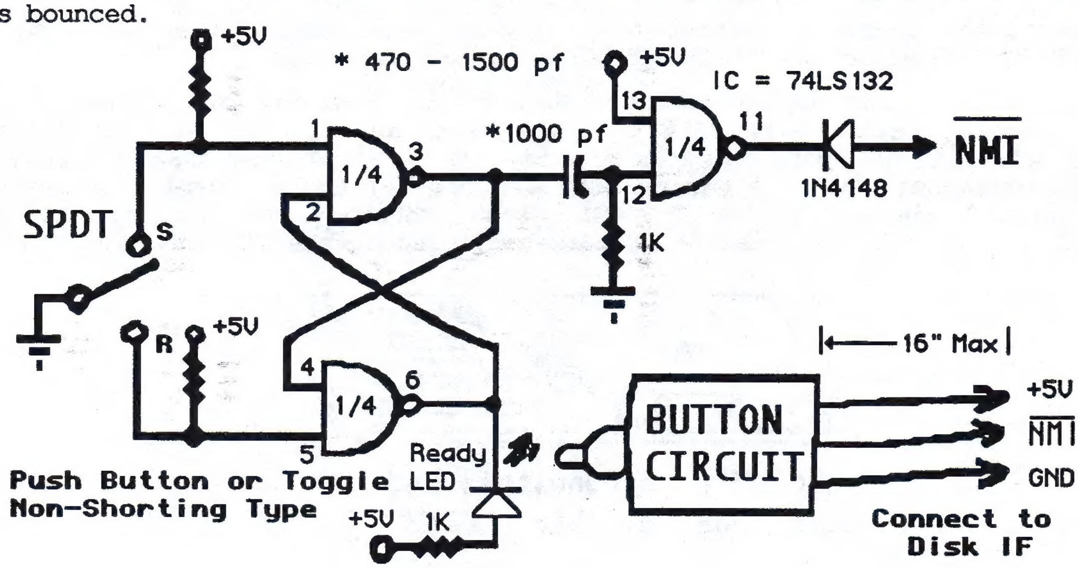
Modified for LKDOS LROS Socket
(This is EPROM compatible)



CIRCUIT for NMI SAVE Push Button - *(AERCO, Ramex or Oliger Only)*
For the NMI save to work properly a push button circuit with no bounce and a short pulse is critical This circuit is only a simple idea but you should use a flip flop or a latch like this.

A capacitor, schmitt trigger type de-bounce circuit shouldn't be used because it doesn't produce a single pulse.

To test the NMI button you can trigger it within a program and the press enter the tune should only play once (start and end) and the program should continue. If the tune repeats its self or the program crashes then the switch has bounced.



OLIGER VERSION LKDOS Installation

Using the LKDOS cartridge with your Oliger disk system will give you extra commands and let you be compatible with all other TS2068s running LKDOS.

Insert the LKDOS cartridge into the cartridge port and turn on the TS2068. Some TS2068s will not initialize when the LKDOS and SAFE boards are used together. If the TS2068 seems to 'lock up' on power up, you will need to modify either the LKDOS or the SAFE board as described below.

The reason for the problem is that both boards are trying to initialize at once. Both boards have a reset circuit that makes them become active at power up.

If you have a power up problem when the LKDOS is installed, you can add a disable switch to the LKDOS cartridge.

To add a disable switch, connect a small wire to the unused hole in the PC board near pin 1 on the 74LS74. This hole is between the notched end of the 74LS74 and the 470 ohm resistor next to it. Carefully scrape the green solder mask off the pad and solder the wire to it. You can add a miniature switch to ground this wire to disable the LKDOS. The ground trace runs around the edge of the solder side of the cartridge.

If you turn on the TS2068 with the switch disabled, you will have to initialize to cartridge with the command RAND USR 96 after power up.

By the way, you don't need to use RAND USR 100: OPEN #4,"dd" when running LKDOS, just put RAND USR 100: in front of all commands.

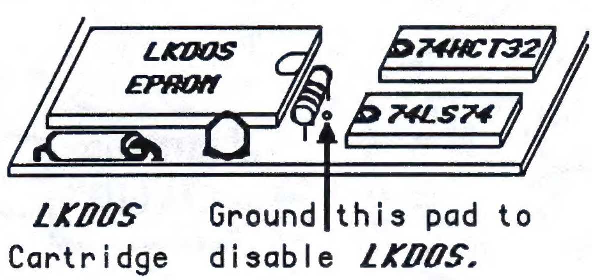
An alternate method of fixing the initialization problem is to disable the power-on reset circuitry on the SAFE board. This way is really better because the LKDOS cartridge has special software to initialize the SAFE board after it has initialized itself on power up.

The easiest way to do this is to remove the 74HCT74 on the SAFE board from its socket and bend pin 1 out. Put the IC back into its socket and wire pin 1 to +5V. Pin 14 on the same IC has +5V on it so you can just connect these two together with some fine wire. If you don't want to solder it, just use a few turns of wire on each pin.

Now, if you turn on the TS2068 while pressing 'i', the LKDOS will initialize the SAFE board after it has initialized itself. If you load an AUTOSTART program with LKDOS though, it will not initialize SAFE. Use the Restore/ command in SAFE to initialize SAFE.

When using the snapshot button, only have the DOS enabled that you want to respond to the NMI as the LKDOS can also perform NMI saves.

If you are using the LKDOS with the SAFE board enabled, you shouldn't use LKDOS commands that use PRINT #4: in them such as windows, printer driver, sequential-files or other commands preceded by PRINT #4 (use RAND USR 100: to precede commands). This is because the vectors for these commands share the same memory patch area as the JLO NMI vector and they will cause the JLO NMI to conflict. When using these LKDOS commands, disable the SAFE board.

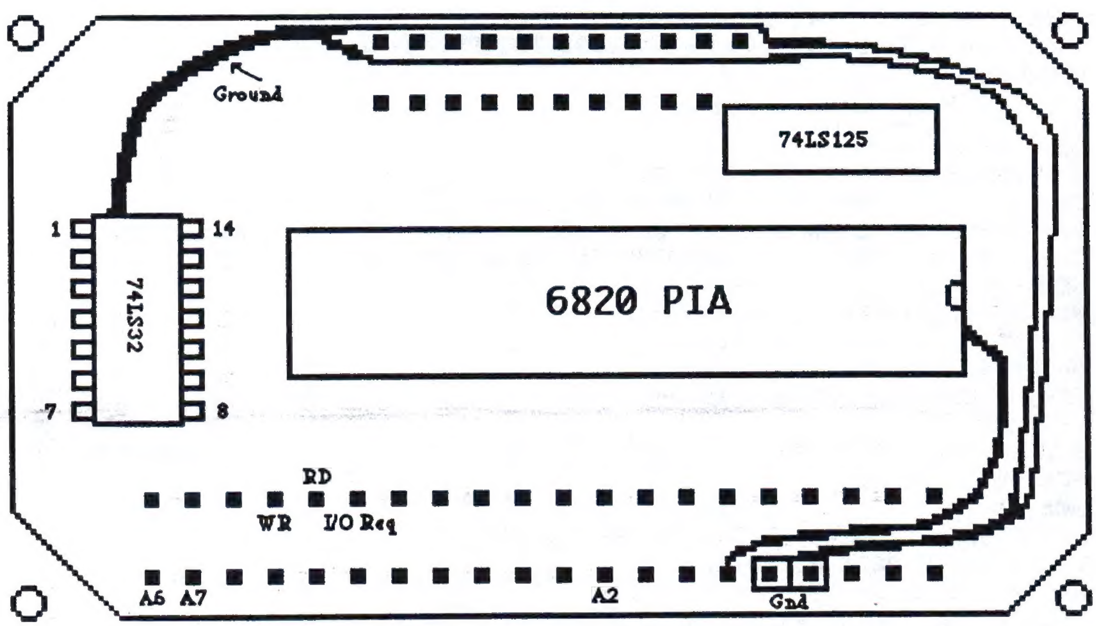


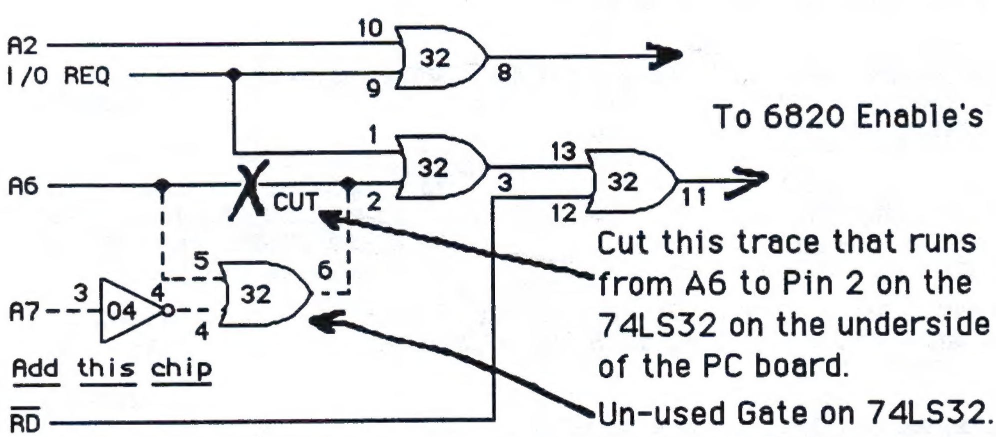
Making the Tasman 'B' CPI Work With the LARKEN L3 & L3F

To make the Tasman version 'B' Centronics Parallel Interface (CPI) work with the LARKEN disk System, acquire a 74LS04 quad NAND gate and some small gauge insulated hook-up wire. Open the Tasman 'B' CPI by first carefully peeling back the plastic label away from the screws. (If you pick up the ends carefully with an X-ACTO knife, the label will press back over the screws when you are done without even a wrinkle.)

Remove the four screws that hold the case together and open the case. Examine the PC board to be certain that you have a 'B' version Tasman CPI. It is a 'B' if it is labeled 'VERSION B'.

Now follow the instructions on the drawing. When you are finished, carefully check all connections against the drawing. Close the case and press the plastic label back into place.





- Bend all of the pins (except pins 4, 7 and 14) on the 74LS04 up, and stack it on top of the 74LS32. Solder pins 4, 7 and 14 of both ICs together.

- Connect A6 on the connector to pin 5 on the 74LS32 with a wire.
- Connect A7 on the connector to pin 3 on the 74LS04 with wire.
- Connect pin 2 on the 74LS32 to pin 6 on the 74LS32.

EXTENDED BASIC DEMO PROGRAMS

```
 10 REM **** WINDOW DEMO ****
 20 RANDOMIZE USR 100: OPEN #4,"dd"
 30 CLS : INK 7: PAPER 0
 40 PRINT #4: OPEN #5,"w0"
 50 PRINT #4: INPUT #0,3,3,15,10
 60 INK 0: PAPER 4
 70 PRINT #4: OPEN #6,"w1"
 80 PRINT #4: INPUT #1,12,10,30,20
 90 INK 0: PAPER 7: CLS
100 PRINT #4: CLEAR 0:PRINT #4: CLEAR 1
110 PRINT #0;"Press Enter for 'SCROLL ??'"
120 LIST #5: LIST #6: LIST #5: LIST #6: LIST #5
```

```
 10 REM **** DRAW DEMO  ****
 20 RANDOMIZE USR 100: OPEN #4,"dd"
 30 LET x=INT(RND*200): LET y=INT(RND*150)
 40 PLOT x,y: PRINT #4: DRAW 40,40,1
 50 PLOT x+2,y+2: PRINT #4: DRAW 36,36,INT(RND*8)+1
 60 GOTO 30
```

```
 10 REM **** FILL DEMO ****
 15 REM --- Draw House ------
 20 RANDOMIZE USR 100: OPEN #4,"dd"
 30 PLOT 30,20: DRAW 70,0: DRAW 60,60: DRAW 0,50: DRAW -35,35
 40 DRAW -60,-60: DRAW 35,-35: DRAW 60,60: DRAW -60,-60: DRAW 0,-50
 50 DRAW -70,0: DRAW 0,50: DRAW 35,35: DRAW -35,-35: DRAW 70,0
 60 REM --- Fill House ------
 70 PRINT #4: CIRCLE 50,50,8
 80 PRINT #4: CIRCLE 100,100,2
 90 PRINT #4: CIRCLE 130,60,6
100 REM -- UDP for Pattern 10 --
110 RESTORE
120 FOR a=23540 TO 23548
130 READ b: POKE a,b
140 NEXT a
150 DATA 15,240,15,240,15,240,15,240,15,240
160 PRINT #4: CIRCLE 80,80,10
170 REM -- Cut Door --------
180 PLOT 60,20: PRINT #4: DRAW 10,20,1
190 PLOT 61,20: PRINT #4: DRAW 8,19,0
```

FROM:
LARKEN ELECTRONICS
RR #2 NAVAN ONTARIO
CANADA K4B-1H9

AUTOSTART PROGRAMS

```
 10 REM POKE into locations 26715 to 26722 - 205,102,0,62,3,211,244,201
 20 CLEAR 65535: REM Works only on pure LARKEN systems
 30 REM RANDOMIZE USR 100: OPEN #4, "dd"
 40 RANDOMIZE USR 100: LOAD "Menu.B1"
 50 CLEAR 27577: REM GOTO 50 to AUTO
 60 LET A=FN A(23635)
 70 POKE A+9,IN 244
 80 RANDOMIZE USR A+5: GOTO 20
 90 DEF FN A(x) =PEEK x+256*PEEK x+1
```

```
10 CLEAR 65535: REM Works on all
20 RANDOMIZE USR 100: OPEN #4, "dd"
30 RANDOMIZE USR 100: LOAD "Menu.B1"
40 CLEAR 27577: REM GOTO 40 to AUTO
50 RANDOMIZE USR 102: RUN
```

NMI-F PROGRAM

```
 10 REM NMI-F.B1 V1.2 The 128 at the end of line 30 is the drive number to go
 20 RESTORE : CLS : PRINT AT 10,5;"NMI-F REBOOT ";            to upon boot-up.
 30 LET c=62+256*128
 40 FOR a=16100 TO 16164 STEP 2: READ b
 50 RANDOMIZE USR 100: POKE a,b: NEXT a
 60 DATA 52723,98,c,818,32,0,0,6945,4415,8226,2561,60672,16048,12811,8194,
50893,10752,8316,13090,10784,8326,12578,52512,201,25662,51707,0,16640,21589,
21327,16724,21586,32
 70 RANDOMIZE USR 100: POKE 8214,16100
 80 PRINT "ACTIVATED": FOR I=1 TO PI: BEEP .042,50: NEXT I
 90 RESTORE: GO TO 110
100 RANDOMIZE USR 100: SAVE "NMI-F.B1" LINE 20
110 RANDOMIZE USR 100: LOAD "Menu.B1"
```